Model Recovery: Models A and B

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1 Question

On the real data, PSIS-LOO strongly prefers Model B (hierarchical over sessions) to Model A (pooled over sessions): elpd_{loo} is -3344.0 for A and -3180.0 for B, a difference of 164.1 with standard error 19.2. Before reading this as evidence that the monkey's psychometric curves vary from session to session, two things need checking:

1. Can LOO identify the model that generated a dataset of this size and design?
2. Could B win on the real data simply because it is more flexible? B contains A as the special case of no between-session variation, so on data from A it can fit at least as well in-sample.

This report answers both with a model recovery study: synthetic datasets are generated from each model, both models are fit to each, and LOO picks one.

2 Models

Both models describe each binary response with the psychometric function

$$\psi(x) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x)$$

for each of the four groups (hand $\times$ unimanual/bimanual).

- **Model A** has one set of $(\beta_0, \beta_1, \gamma_h, \gamma_l)$ per group, shared by all sessions (16 parameters).
- **Model B** draws a set per group and session from group-level distributions with means and between-session sds (8 hyperparameters per group, plus $4 \times 4 \times 42$ session-level parameters).

The model structures and priors are described in the reports "Convergence Simulations Model A" and "Convergence Simulations Model B"; the priors were not changed for this study.

3 Design

Every figure and number in this report is produced by `Model_Recov_BA.py`. Its cell "Numbers for Model_Recov_BA_Report.tex" prints the numbers that `Model_Recov_Results.txt` and the figure do not show.

Replicates. Each of 10 replicates draws two fresh datasets on the real trial design (the real stimuli, groups and sessions, 12937 trials in 42 sessions):

- **data A**: responses from Model A, with one parameter set per group;
- **data B**: new session-level parameters drawn from Model B's hyperparameters, then responses from those.

True values. To make the synthetic data resemble the real data, the true values are the posterior means of each model's fit to the real data (Table 1). They only generate data; the priors do not use them.

Table 1: True values. Top: Model A's group parameters. Bottom: Model B's hyperparameters, with the lapse-rate ones on the logit scale.

	left_uni	left_bi	right_uni	right_bi
β_0	-1.422	-0.156	-0.967	0.608
β_1	4.588	3.956	3.980	3.982
γ_h	0.002	0.041	0.003	0.045
γ_l	0.009	0.016	0.006	0.012
μ_{β_0} / σ_{β_0}	-1.487 / 0.763	-0.099 / 0.772	-1.056 / 1.011	0.686 / 0.759
μ_{β_1} / σ_{β_1}	4.968 / 0.285	4.321 / 1.266	4.580 / 0.173	4.340 / 0.615
μ_{γ_h} / σ_{γ_h}	-5.203 / 0.359	-2.350 / 0.591	-4.729 / 0.371	-1.816 / 0.362
μ_{γ_l} / σ_{γ_l}	-3.445 / 0.373	-3.442 / 0.393	-3.717 / 0.381	-3.228 / 0.431

Fits and comparison. Both models are fit to both datasets of every replicate (4 chains of 1000 draws after 1000 tuning steps; about 1 minute per Model A fit and 3 minutes per Model B fit). For each dataset,

$$\Delta = \mathrm{elpd}_{\mathrm{loo}}(\mathrm{B}) - \mathrm{elpd}_{\mathrm{loo}}(\mathrm{A}),$$

so $\Delta > 0$ favors B. Its standard error dse is computed from the trial-wise differences, as `az.compare` does, and $z = \Delta/\mathrm{dse}$. The model with the higher $\mathrm{elpd}_{\mathrm{loo}}$ counts as selected.

4 Results

Table 2: Which model LOO selects (the higher $\mathrm{elpd}_{\mathrm{loo}}$), over 10 replicates.

	selects A	selects B
data from A	9	1
data from B	0	10

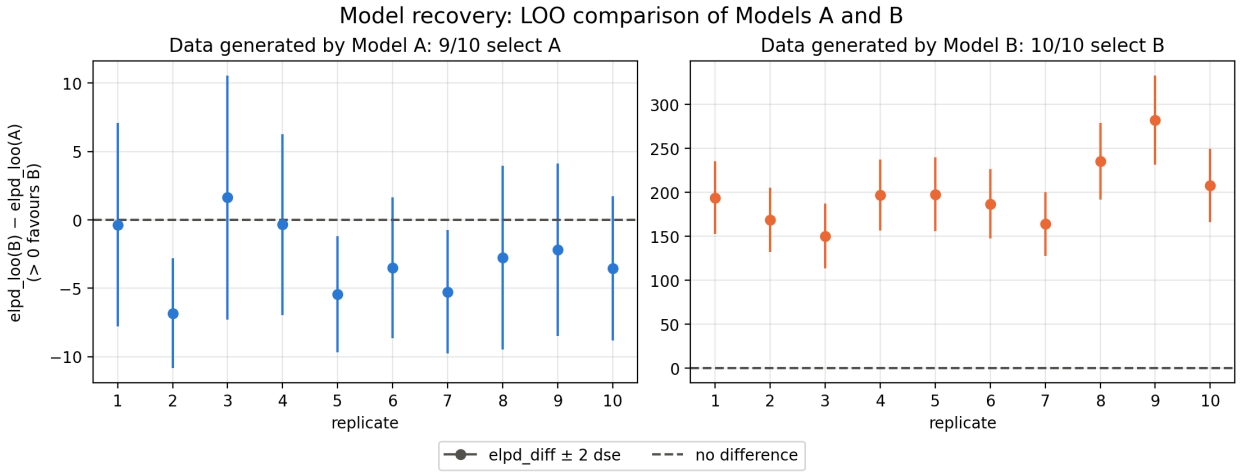


Figure 1: $\Delta = \text{elpd}_{\text{loo}}(\text{B}) - \text{elpd}_{\text{loo}}(\text{A})$ with ± 2 dse for every replicate. Left: data generated by Model A. Right: data generated by Model B. Note the different vertical scales.

Data from B. LOO selects B every time, by a wide margin: Δ ranges from 150.2 to 282.4 (median 195.4) and z from 8.16 to 11.12. Model B's effective number of parameters p_{loo} is 166.7–189.6, against 12.5–14.7 for A.

Data from A. LOO selects A in 9 of 10 replicates, but by small margins: Δ ranges from -6.83 to 1.63 (median -3.14), with z from -3.41 to 0.36. The difference is more than 2 dse in favor of A in only 3 of the 10 replicates; in the other 7, $|z| \leq 2$. The one replicate where B is selected is a near-tie ($\Delta = 1.63$, $z = 0.36$). This is the expected pattern when the true model is the simpler one nested inside the other: B can mimic A by shrinking its between-session sds toward 0, and then loses only the small LOO cost of its extra flexibility. Accordingly, B's p_{loo} on data from A is 53.9–83.6, much lower than on data from B. The comparison is therefore asymmetric: it identifies session-to-session variation clearly when it is there, but it cannot strongly confirm that it is absent.

Real data. The real difference, $\Delta = 164.1$ ($z = 8.55$), lies inside the range produced by data from B (150.2 to 282.4) and far outside the range produced by data from A (-6.83 to 1.63). Model B's p_{loo} on the real data, 178.7, also matches the data-from-B range (166.7–189.6) rather than the data-from-A range.

5 Diagnostics

Sampling. All 40 fits sampled without divergences. Over all fits, the maximum $\hat{R}$ is 1.004 for Model A and 1.009 for Model B, and the minimum bulk ESS is 2411 and 857.

These $\hat{R}$ and ESS values leave out the variables z_σ behind Model B's between-session sds of β_0 and the lapse rates, which enter the model only through an absolute value ($\sigma_{\beta_0} = |2z_{\sigma_{\beta_0}}|$, and similarly for σ_γ). Their sign is therefore not identified, and when the true sd is clearly above 0 different chains settle on opposite signs. On data from B, their $\hat{R}$ is 1.53–1.74, which looks like a failed fit. Nothing that enters the likelihood depends on that sign, so the sds themselves, and everything computed from them, are checked instead (the values above). On data from A, where the sds are close to 0 and the two signs overlap, their $\hat{R}$ is at most 1.059.

LOO. The largest Pareto k over all fits is 0.645, below the 0.7 threshold above which PSIS-LOO estimates are unreliable. No fit has a k above 0.7.

6 Conclusions

- With the real design and data of this size, LOO identifies session-to-session variation reliably: when it is present at the level estimated from the real data, B wins every time with $z > 8$.
- B's flexibility alone does not make it win. On data without session variation, B loses or ties (Δ at most 1.63), so B's advantage on the real data, 164.1, is not something its extra parameters produce by themselves.
- The real-data comparison looks like the data-from-B replicates in both Δ and p_{loo}. Taken together, the preference for Model B on the real data is good evidence that the psychometric curves vary between sessions.
- The reverse question, confirming that some form of session variation is absent, is answered only weakly by LOO, since the more flexible model loses by just a few elpd points when the simpler one is true.

Limitations.

1. Ten replicates per generating model: enough to see that the two distributions of Δ are far apart, but not to estimate an error rate precisely.
2. Data from B use one set of hyperparameters, those of the real fit. With smaller between-session sds, the gap between the models would shrink.
3. Only Models A and B are compared. Model C (distractor amplitude as a covariate) is not part of this study.